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Blueprint based exit material for

AI & big data

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❖ Knowledge representation of AI

● Following are the kind of knowledge which needs to be represented in AI systems:

- **Object:** All the facts about objects in our world domain. E.g., Guitars contains strings, trumpets are brass instruments.
- **Events:** Events are the actions which occur in our world.
- **Performance:** It describes behaviour which involves knowledge about how to do things.
- **Meta-knowledge:** It is knowledge about what we know.
- **Facts:** Facts are the truths about the real world and what we represent.
- **Knowledge-Base:** The central component of the knowledge-based agents is the knowledge base. It is represented as KB. The Knowledgebase is a group of the Sentences (Here, sentences are used as a technical term and not identical with the English language)

● AI knowledge cycle

- An Artificial intelligence system has the following components for displaying intelligent behaviour:
 - Perception
 - Learning
 - Knowledge Representation and Reasoning
 - Planning
 - Execution

● Techniques of knowledge representation

- There are mainly four ways of knowledge representation which are given as follows:
 1. Logical Representation
 2. Semantic Network Representation
 3. Frame Representation
 4. Production Rules

❖ Demonstrate solving problem by searching

The process of problem-solving using searching consists of the following steps.

- Define the problem
- Analyse the problem
- Identification of possible solutions

- Choosing the optimal solution
- Implementation

Problem-solving agents in AI mostly used these search strategies or algorithms to solve a specific problem and provide the best result. Problem-solving agents are the goal-based agents and use atomic representation.

❖ Properties of searching algorithm

- Completeness - it guarantees to return a solution if at least any solution exists for any random input.
- Optimality - is guaranteed to be the best solution (lowest path cost) among all other solutions.
- Time complexity – measure of time for an algorithm to complete its task.
- Space complexity - it is the maximum storage space required at any point during the search, as the complexity of the problem
- There are d/t types of searching algorithm. Thus

A. Uniformed(blind)

- Does not contain any domain knowledge such as closeness, the location of the goal
- Operates in brute force-way
- Do not have additional information about state or search space other than how to traverse the tree.
- Blind Search is also known as uninformed search, and it does not contain any domain information such as closeness, location of the goal, etc. Hence the most appropriate situation that can be used for the blind search is Small-search Space.

It can be divided into five main types:

i. Breadth-first search

- Performed by exploring all nodes at a given depth before proceeding to the next level.
- It uses First-in First-out (FIFO) strategy as it gives the shortest path to achieving the solution.
- It generally starts from the root node and examines the neighbor nodes and then moves to the next level.
- Is used where the given problem is very small and space complexity is not considered.

ii. Uniform cost search

- Is considered the best search algorithm for a weighted graph or graph with costs. It searches the graph by giving maximum priority to the lowest cumulative cost. Uniform cost search can be implemented using a priority queue.

- This algorithm is optimal as the selection of paths is based on the lowest cost.
- The algorithm does not consider how many steps it goes to reach the lowest path. This may result in an infinite loop also.
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iii. Depth-first search

- Require less memory and less time complexity, do not require much more search, doesn't give guarantee to give solution.
- Uses Last-in, First-out (LIFO) strategy and hence it can be implemented by using stack.
- Uses backtracking. That is, it starts from the initial state and explores each path to its greatest depth before it moves to the next path.

iv. Iterative deepening depth-first search

- is a combination of depth-first search and breadth-first search
- find the best depth limit by gradually adding the limit until the defined goal state is reached.
- It offers fast search and uses memory efficiently.
- It does all the works of the previous stage again and again

v. Bidirectional Search

- Is completely different from all other search strategies. It executes two simultaneous searches called forward-search and backwards-search and reaches the goal state. Here, the graph is divided into two smaller sub-graphs. In one graph, the search is started from the initial start state and in the other graph; the search is started from the goal state. When these two nodes intersect each other, the search will be terminated.
- Bidirectional search requires both start and goal state to be well defined and the branching factor to be the same in the two directions.
- This algorithm searches the graph fast.
- It requires less memory to complete its action.
- The goal state should be pre-defined.
- The graph is quite difficult to implement.

State space search – most straightforward approach for planning algorithm

B. Informed/heuristic search/ directed search

- use domain knowledge, require detail
- Problem information is available which can guide the search.
- can find a solution more efficiently than an uninformed search strategy

- A way which might not always be guaranteed for best solutions but guaranteed to find a good solution in reasonable time.
- Is used to achieve the goal state with the lowest cost possible. This function estimates how close a state is to the goal.

It can be divided into five main types:

i. Greedy search / **Best First Search Algorithm**

- Uses the properties of both depth-first search and breadth-first search. Greedy best-first search traverses the node by selecting the path which appears best at the moment. The closest path is selected by using the heuristic function.

ii. **A* search**

- Is a combination of both uniform cost search and greedy best-first search algorithms? It uses the advantages of both with better memory usage. It uses a heuristic function to find the shortest path. A* search algorithm uses the sum of both the cost and heuristic of the node to find the best path.
- This algorithm can be used to solve very complex problems also it is an optimal one.
- The A* search is based on heuristics and cost. It may not produce the shortest path.
- The usage of memory is more as it keeps all the nodes in the memory

❖ Understand intelligent agents and type

The agents sense the environment through sensors and act on their environment through actuators. An AI agent can have mental properties such as knowledge, belief, intention, etc.

An agent can be anything that perceive its environment through sensors and act upon that environment through actuators. An Agent runs in the cycle of **perceiving, thinking, and acting**. An agent can be:

- **Human-Agent:** A human agent has eyes, ears, and other organs which work for sensors and hand, legs, vocal tract work for actuators.
- **Robotic Agent:** A robotic agent can have cameras, infrared range finder, NLP for sensors and various motors for actuators.
- **Software Agent:** Software agent can have keystrokes, file contents as sensory input and act on those inputs and display output on the screen.

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Sensor: Sensor is a device which detects the change in the environment and sends the information to other electronic devices. An agent observes its environment through sensors.

Actuators: Actuators are the component of machines that converts energy into motion. The actuators are only responsible for moving and controlling a system. An actuator can be an electric motor, gears, rails, etc.

Effectors: Effectors are the devices which affect the environment. Effectors can be legs, wheels, arms, fingers, wings, fins, and display screen.

An intelligent agent is an autonomous entity which acts upon an environment using sensors and actuators for achieving goals. An intelligent agent may learn from the environment to achieve their goals. A thermostat is an example of an intelligent agent.

❖ Rational agent

Rational agents in AI are very similar to intelligent agents. The task of AI is to design an agent program which implements the agent function. The structure of an intelligent agent is a combination of architecture and agent program. It can be viewed as:

Agent = Architecture + Agent program

❖ Types of AI Agents

- Agents can be grouped into five classes based on their degree of perceived intelligence and capability. All these agents can improve their performance and generate better action over the time. These are given below:

1. Simple Reflex Agent

- It follows the Condition-action rule.
- Only succeed in the fully observable environment.
- Take decisions on the basis of the current percepts and ignore the rest of the percept history.

2. Model-based reflex agent

- Consider the percept history in their actions.

3. Goal-based agents - The knowledge of the current state environment is not always sufficient to decide for an agent to what to do.
4. Utility-based agent - deals with the happy and unhappy state
5. Learning agent

❖ Fundamental of artificial intelligence programming technique

- Machine learning
- Deep learning
- Automation and robotics

❖ Social technical and ethical uses of machines with AI

Big data

❖ Describe big data and use case from selected business domain

- Big data is a field that treats ways to analyse systematically extract information from or otherwise deal with data sets. Data can be large or complex to be dealt with by traditional data processing applications software
 - A large amount of data
 - It is a popular term used to express the exponential growth of data.
 - Big data is difficult to store, collect, maintain, analyse and visualize.
 - **Big data tools:** Apache Hadoop, Apache Storm, Cassandra, Mongo DB, Neo4j.
 - **Big data sources:** Amazon, Redshift, Mongo DB

● Challenges of big data:

- Uncertainty of data management
- The talent gap in big data
- Getting data into a big data structure
- Synchronizing across data sources
- Integration

● Benefits of big data:

- Cost
- Time reduction

- Speeding up decision making
- Analyse in real-time
- Model and Test variation

● Characteristics of big data:

- Volume
- Velocity
- Variety

Types of big data:

- Structured
- unstructured
- Semi-structured
- hybrid

Use cases of big data

- Recommendation engine
- Analyzing call detail records
- Fraud detection
- Market basket analysis
- sentiment analysis

Data in petabyte size is known as big data.

❖ Explain NoSQL big data management

NoSQL is a set of concepts that allows the rapid and efficient processing of data sets with a focus on performance, reliability, and agility.

NoSQL IS

- *It's more than rows in tables*
- *It's free of joins*
- *It's schema(ERM)-free*
- *It works on many processors*
- *It uses shared-nothing commodity computers*
- *It supports linear scalability*

- *It's innovative*

NoSQL IS NOT

- *It's not about the SQL language*
- *It's not only open source*
- *It's not only big data*
- *It's not about cloud computing*
- *It's not about a clever use of RAM and SSD*
- *It's not an elite group of products*

Here are some essential benefits of using NoSQL databases-

1. **Scalability-** NoSQL databases are designed to be easily vertically/horizontally scalable by adding more nodes or expanding the infrastructure. This makes them a good fit for large-scale applications requiring high throughput and low latency.
2. **Flexibility-** NoSQL databases can handle unstructured and semi-structured data formats, which makes them more flexible than SQL databases that require a predefined schema and rigid data structure. NoSQL systems can adapt to flexible data models and requirements more easily.
3. **Performance-** NoSQL databases are optimized for fast reading and writing, which can be especially beneficial for real-time applications, such as gaming, social media, or e-commerce. NoSQL databases can handle vast data and complex queries without affecting performance.
4. **Availability-** NoSQL databases are highly available and fault-tolerant, with built-in replication and sharding capabilities. NoSQL databases can continue functioning even if some nodes or servers fail without losing data or experiencing downtime.
5. **Cost-Effectiveness-** NoSQL databases are more cost-effective than SQL databases, especially for large-scale applications that require a lot of hardware and infrastructure. NoSQL databases can run on commodity hardware and open-source software, which can reduce licensing and maintenance costs.

Here are some of the disadvantages of NoSQL databases-

1. **Limited Query Capabilities-** NoSQL databases are optimized for specific use cases and data formats, meaning they may not support complex queries, data relationships, and SQL databases. This can make performing ad-hoc or exploratory data analysis on NoSQL databases challenging.
2. **Lack Of Standardization-** NoSQL databases do not have a standard query language or data model, making it harder to migrate data or integrate with other systems. Each NoSQL database has its syntax and semantics, making it harder for developers to learn and use.
3. **Inconsistent Data Quality-** NoSQL databases do not enforce strict data types or constraints, meaning data quality and consistency may be more difficult to ensure with SQL databases. This can lead to data duplication, inconsistencies, or errors, especially in large and distributed systems.
4. **Limited Transaction Support-** NoSQL systems may not support full ACID transactions, meaning they may not guarantee consistency, isolation, and durability of data changes in all situations. This can make it harder to ensure data integrity and recover from failures.
5. **Lack Of Mature Ecosystem-** NoSQL databases are relatively new compared to SQL databases, meaning they may not have as mature an ecosystem of tools, libraries, and technologies. This can make finding support, documentation, and best practices for specific use cases and applications harder.

❖ Concept and applications of Data mining and warehousing

Data mining – fetch important data from database

Its application include

- Financial Data Analysis
- Retail Industry
- Telecommunication Industry
- Biological Data Analysis
- Other Scientific Applications

- Intrusion Detection

Data warehousing

- Organizing and compiling data into one database
- Is the technology that collects the data from various sources within the organization to provide meaningful business insights.
- Is designed for the analysis of data rather than transaction processing.
- Data warehousing involves data cleaning, data integration, and data consolidations

Application of data warehousing

- E – Commerce
- Retail
- AI/ML
- Sustainability and climate action
- Manufacturing and supply chain
- Healthcare
- Banking and finance
- Financial auditing
- Pharmaceuticals

Functions of Data Warehouse Tools and Utilities

The following are the functions of data warehouse tools and utilities –

- **Data Extraction** – Involves gathering data from multiple heterogeneous sources.
- **Data Cleaning** – Involves finding and correcting the errors in data.
- **Data Transformation** – Involves converting the data from legacy format to warehouse format.
- **Data Loading** – Involves sorting, summarizing, consolidating, checking integrity, and building indices and partitions.
- **Refreshing** – Involves updating from data sources to warehouse.

Important Features of Data Warehouse

The Important features of Data Warehouse are given below:

1. Subject Oriented

A data warehouse is subject-oriented. It provides useful data about a subject instead of the company's ongoing operations, and these subjects can be customers, suppliers, marketing, product, promotion, etc. A data warehouse usually focuses on modeling and analysis of data that helps the business organization to make data-driven decisions.

2. Time-Variant:

The different data present in the data warehouse provides information for a specific period.

3. Integrated

A data warehouse is built by joining data from heterogeneous sources, such as social databases, level documents, etc.

4. Non- Volatile

Data Mining	Data Warehousing
Data mining is the process of determining data patterns.	A data warehouse is a database system designed for analytics.
Data mining is generally considered as the process of extracting useful data from a large set of data.	Data warehousing is the process of combining all the relevant data.
Business entrepreneurs carry data mining with the help of engineers.	Data warehousing is entirely carried out by the engineers.
In data mining, data is analyzed repeatedly.	In data warehousing, data is stored periodically.
Data mining uses pattern recognition techniques to identify patterns.	Data warehousing is the process of extracting and storing data that allow easier reporting.
One of the most amazing data mining technique is the detection and identification of the unwanted errors that occur in the system.	One of the advantages of the data warehouse is its ability to update frequently. That is the reason why it is ideal for business entrepreneurs who want up to date with the latest stuff.
The data mining techniques are cost-efficient as compared to other statistical data applications.	The responsibility of the data warehouse is to simplify every type of business data.
The data mining techniques are not 100 percent accurate. It may lead to serious consequences in a certain condition.	In the data warehouse, there is a high possibility that the data required for analysis by the company may not be integrated into the warehouse. It can simply lead to loss of data.
Companies can benefit from this analytical tool by equipping suitable and accessible knowledge-based data.	Data warehouse stores a huge amount of historical data that helps users to analyze different periods and trends to make future predictions.